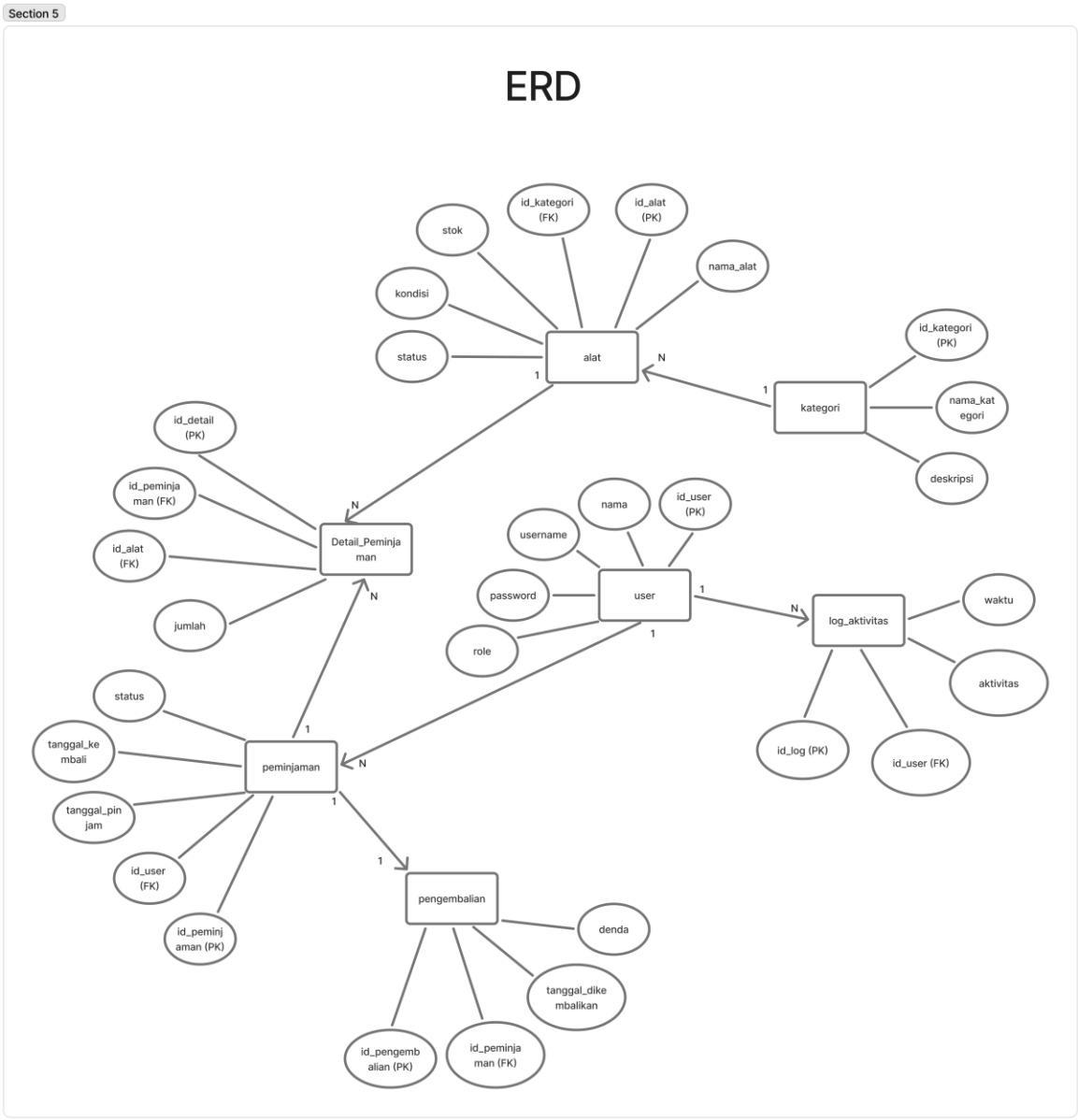
IIS MUTHMAINAH

XII RPL 1

1. Dokumentasi
   a. ERD



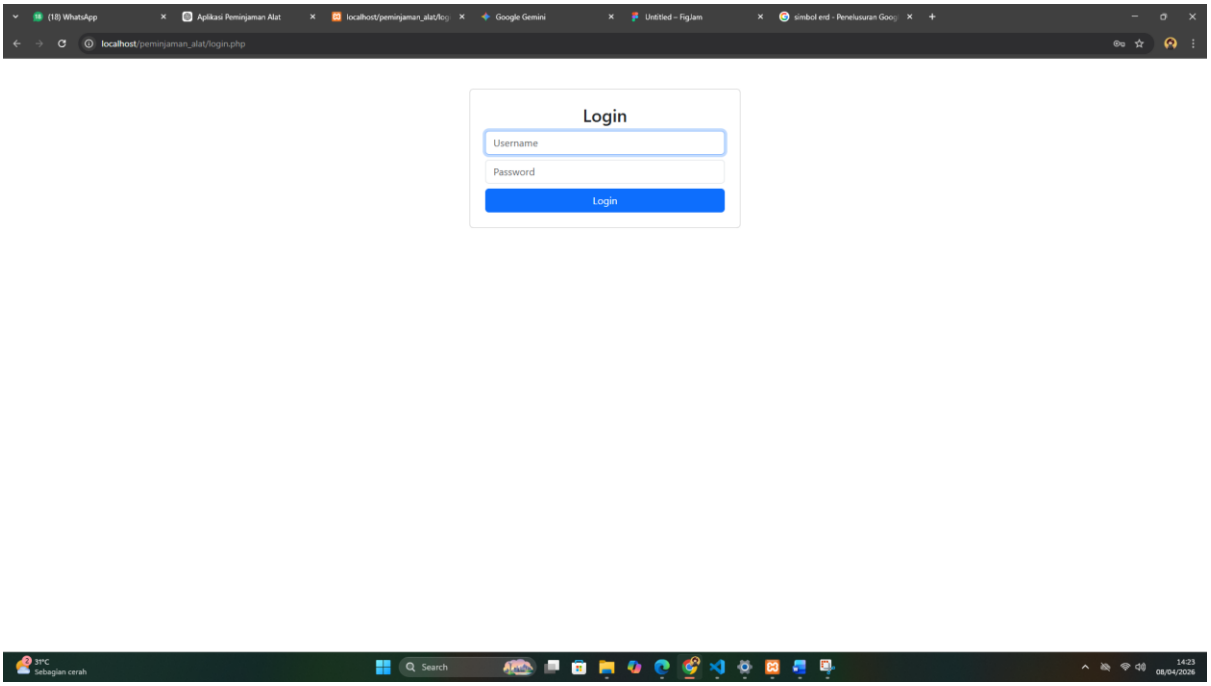
   b. Deskripsi Program

Aplikasi ini digunakan untuk mengelola peminjaman alat berbasis web, dengan 3 role yaitu admin, petugas, dan peminjam. Sistem memungkinkan pengelolaan data alat, transaksi peminjaman, pengembalian, serta laporan.

   c. Dokumentasi Fungsi/Prosedur

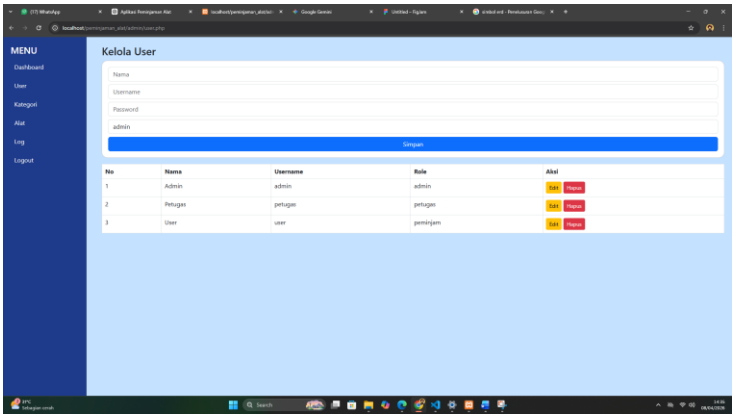
Sistem peminjaman alat memiliki beberapa fungsi utama yang saling terhubung dalam menjalankan proses aplikasi. Fungsi pertama adalah login, dimana

pengguna memasukkan username dan password untuk diverifikasi ke dalam database sebelum dapat mengakses sistem.

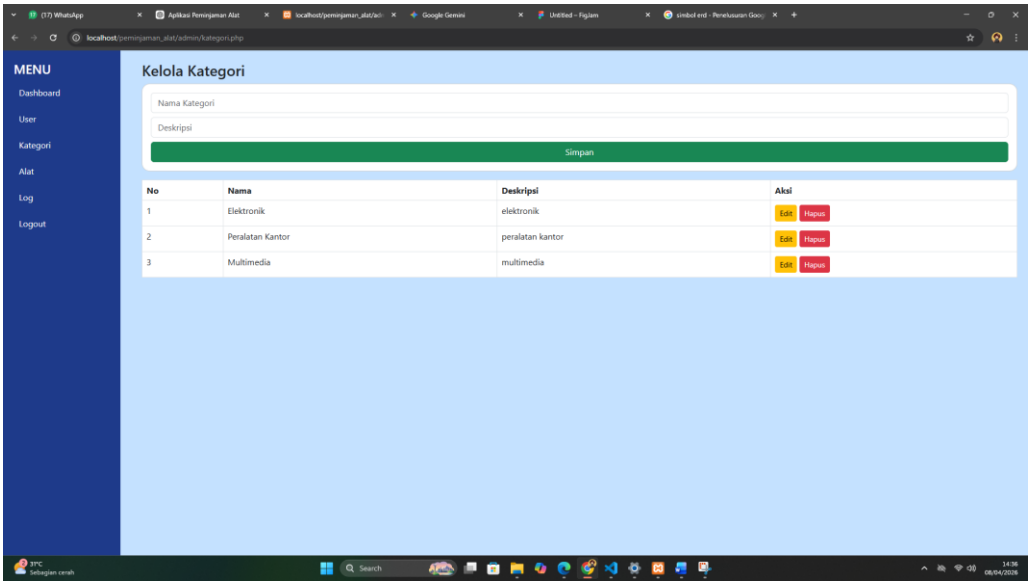


Selanjutnya terdapat fungsi pengelolaan data seperti user, kategori, dan alat yang dilakukan oleh admin. Proses ini menggunakan metode CRUD yaitu menambah, mengubah, menghapus, dan menampilkan data agar sistem tetap terorganisir.

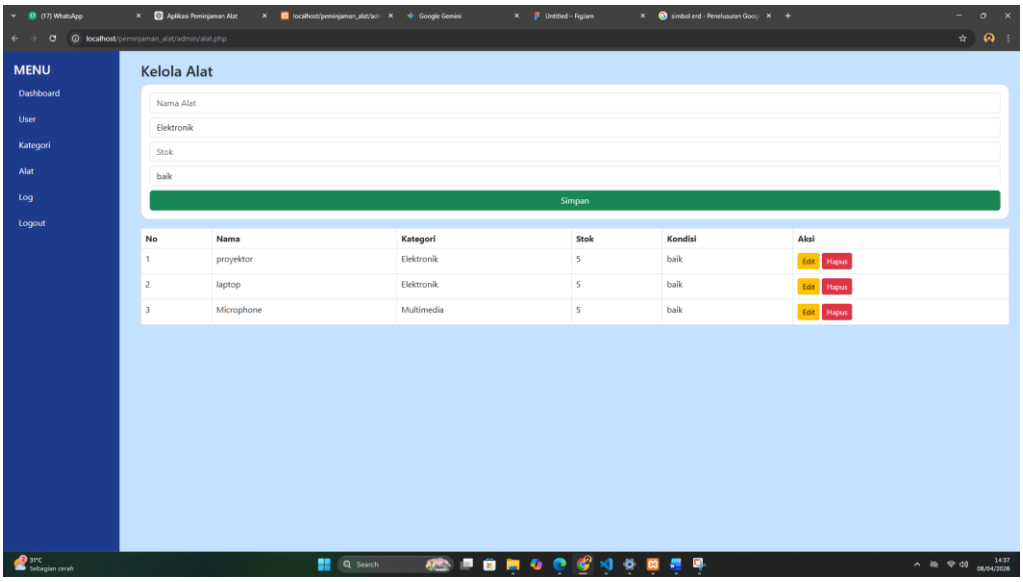
user



Kategori

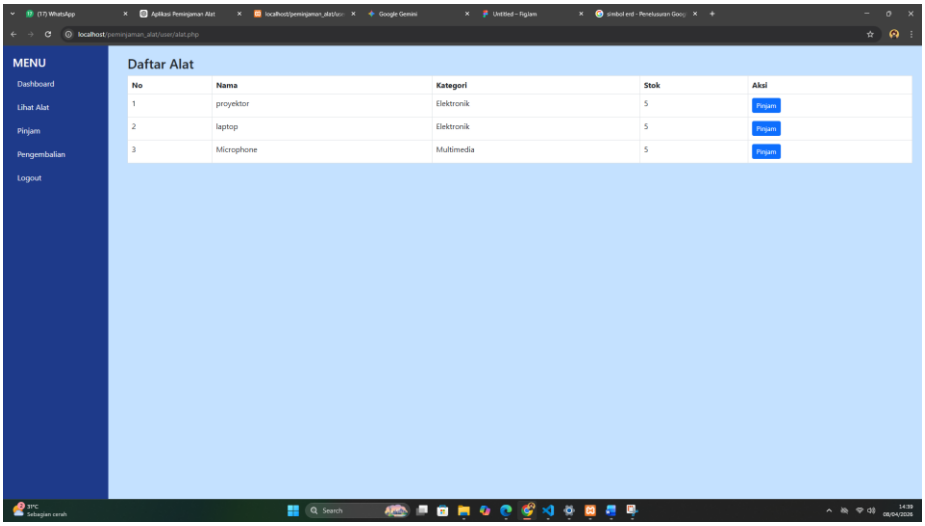


Alat

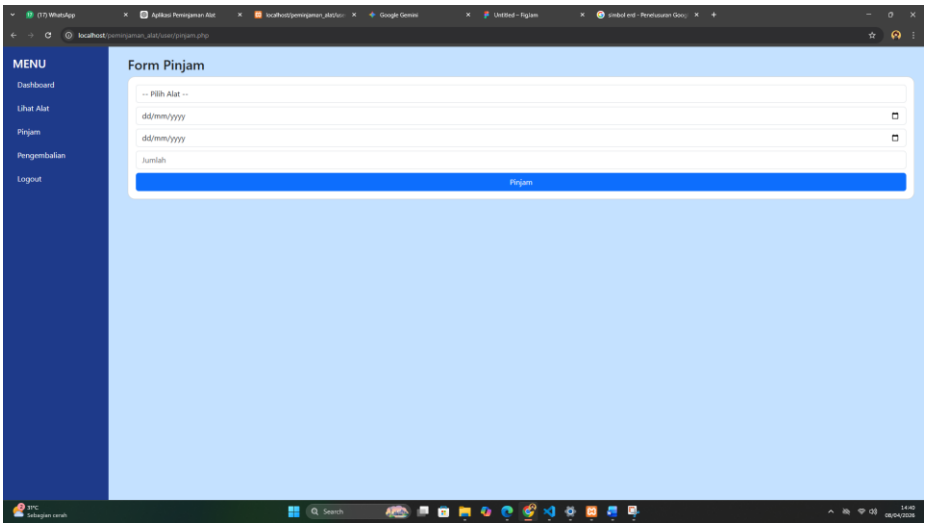


fungsi peminjaman alat yang dilakukan oleh peminjam dengan memilih alat, menentukan tanggal pinjam dan kembali, serta jumlah alat.

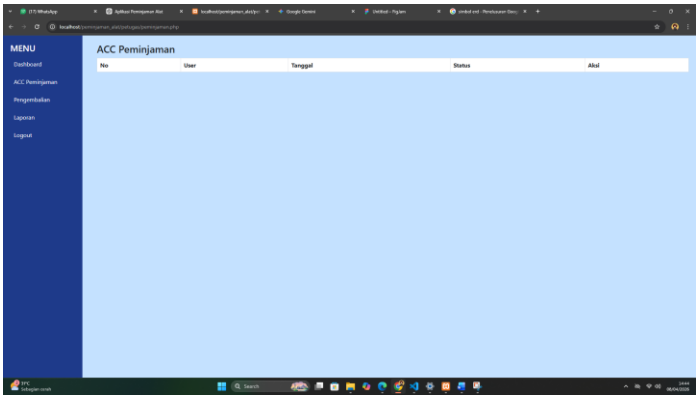
Lihat alat



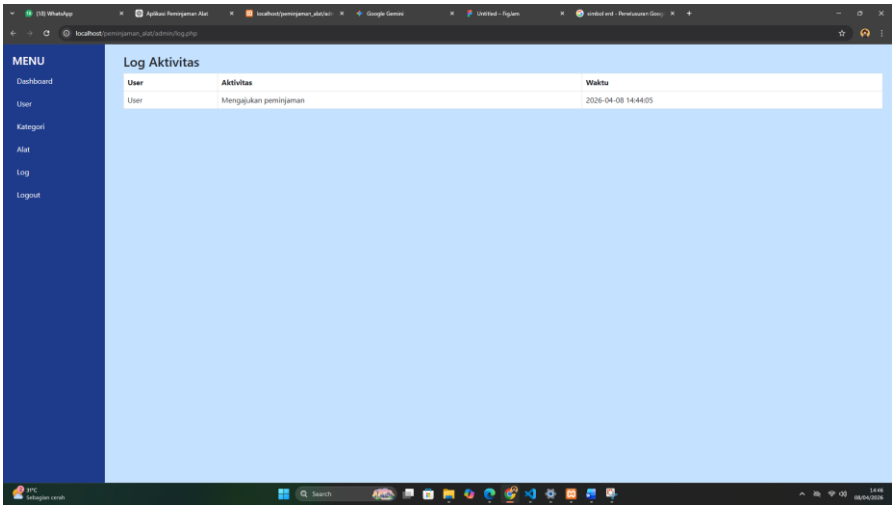
Pinjam



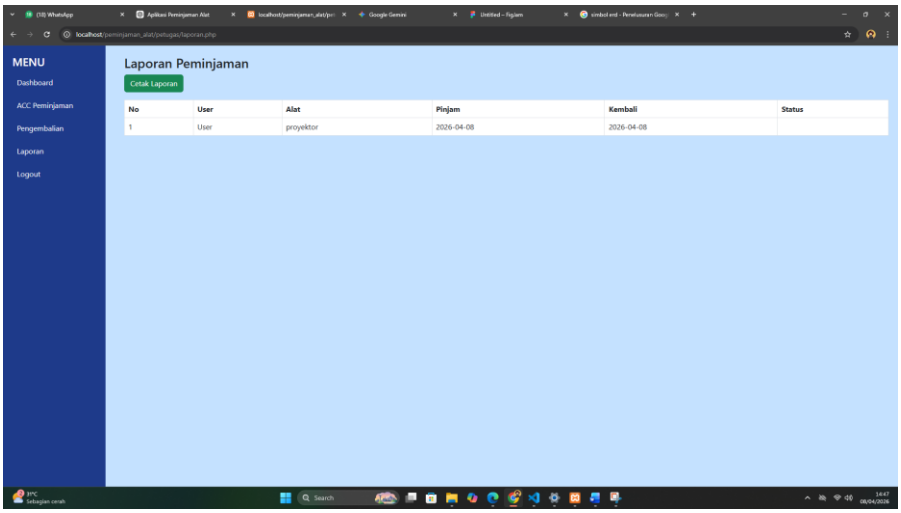
petugas memiliki fungsi untuk menyetujui peminjaman atau ACC terhadap pengajuan yang masuk. Jika disetujui, maka status berubah menjadi dipinjam dan stok alat akan berkurang sesuai jumlah yang digunakan.



Sistem juga dilengkapi dengan fungsi log aktivitas untuk mencatat setiap tindakan pengguna dalam sistem.

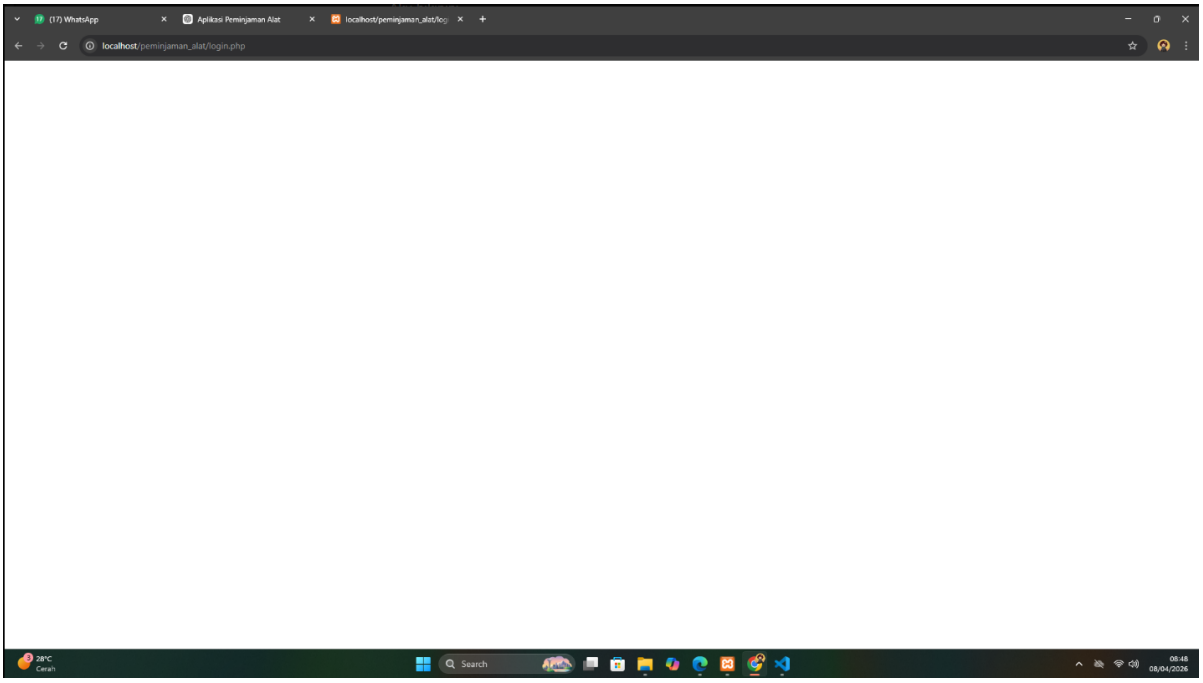


Terdapat fungsi laporan yang digunakan untuk menampilkan dan mencetak data peminjaman serta pengembalian. Laporan ini membantu dalam evaluasi dan dokumentasi kegiatan dalam sistem.

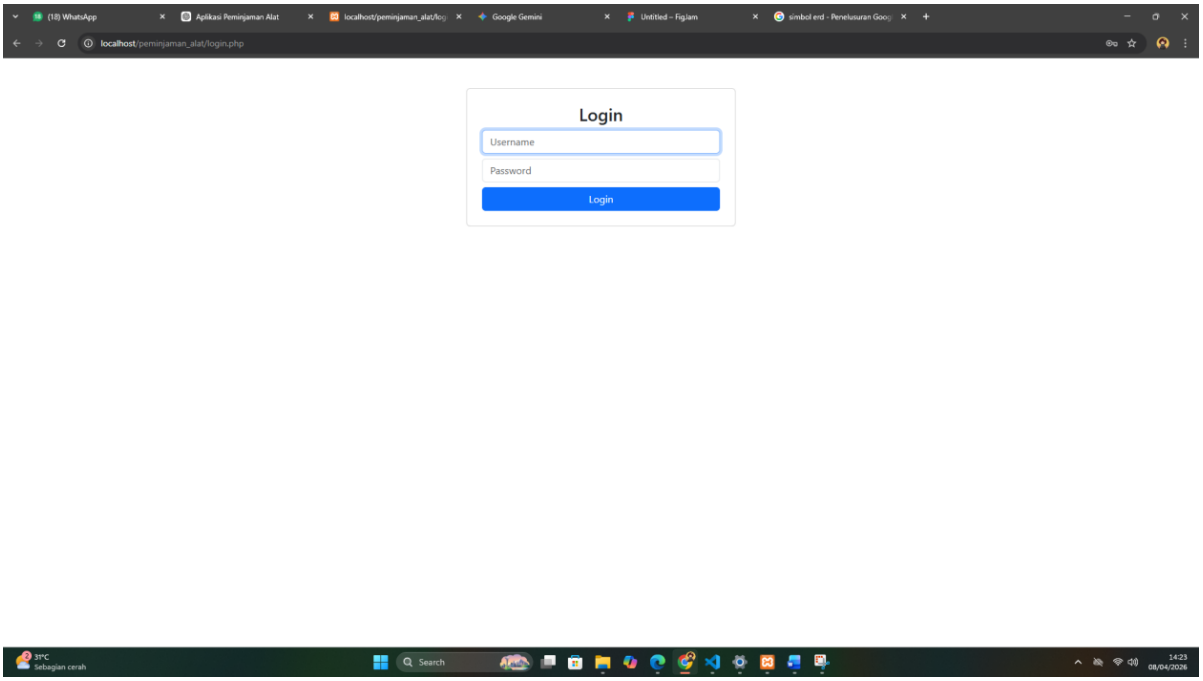


d. Debuging

Pada tahap pengujian sistem, sempat terjadi kendala dimana aplikasi hanya menampilkan layar putih (blank page) saat dijalankan, yang menandakan adanya error pada program. Setelah dilakukan pengecekan, diketahui bahwa penyebabnya adalah file kode yang belum tersimpan sehingga perubahan tidak terbaca oleh sistem. Setelah kode disimpan dengan benar dan dijalankan kembali, aplikasi dapat berjalan normal, sehingga dapat disimpulkan bahwa penyimpanan kode sebelum pengujian sangat penting untuk menghindari error.



Setelah di benarkan



e. Pengujian dan Tangkapan layer hasil uji

admin/alat.php

```php
<?php include '../assets/header.php'; ?>

<?php
if(isset($_POST['simpan'])){
    if($_POST['id']==""){
        mysqli_query($conn,"INSERT INTO alat VALUES(NULL,'$_POST[nama]','$_POST[kategori]','$_POST[stok]','$_POST[kondisi]','tersedia')");
    } else {
        mysqli_query($conn,"UPDATE alat SET
        nama_alat='$_POST[nama]',
        id_kategori='$_POST[kategori]',
        stok='$_POST[stok]',
        kondisi='$_POST[kondisi]'
        WHERE id_alat='$_POST[id]'");
    }
}

if(isset($_GET['hapus'])){
    mysqli_query($conn,"DELETE FROM alat WHERE id_alat='$_GET[id]'");
}

$edit = null;
if(isset($_GET['edit'])){
    $edit = mysqli_fetch_assoc(mysqli_query($conn,"SELECT * FROM alat WHERE id_alat='$_GET[id]'"));
}
?>

<h3>Kelola Alat</h3>

<form method="post" class="card p-3 mb-3">
<input type="hidden" name="id" value="<?= $edit['id_alat'] ?? '' ?>">

<input name="nama" class="form-control mb-2" placeholder="Nama Alat" value="<?= $edit['nama_alat'] ?? '' ?>">

<select name="kategori" class="form-control mb-2">
<?php
$kat = mysqli_query($conn,"SELECT * FROM kategori");
while($k=mysqli_fetch_assoc($kat)){
?>
<option value="<?= $k['id_kategori'] ?>"><?= $k['nama_kategori'] ?></option>
<?php } ?>
</select>
```

```php
<input name="stok" class="form-control mb-2" placeholder="Stok" value="<?= $edit['stok'] ?? '' ?>">

<select name="kondisi" class="form-control mb-2">
<option>baik</option>
<option>rusak</option>
</select>

<button name="simpan" class="btn btn-success">Simpan</button>
</form>

<table class="table table-bordered">
<tr>
<th>No</th><th>Nama</th><th>Kategori</th><th>Stok</th><th>Kondisi</th><th>Aksi</th>
</tr>

<?php
$no=1;
$data = mysqli_query($conn,"SELECT alat.*, kategori.nama_kategori
FROM alat JOIN kategori ON alat.id_kategori=kategori.id_kategori");

while($d=mysqli_fetch_assoc($data)){
?>

<tr>
<td><?= $no++ ?></td>
<td><?= $d['nama_alat'] ?></td>
<td><?= $d['nama_kategori'] ?></td>
<td><?= $d['stok'] ?></td>
<td><?= $d['kondisi'] ?></td>
<td>
<a href="?edit&id=<?= $d['id_alat'] ?>" class="btn btn-warning btn-sm">Edit</a>
<a href="?hapus&id=<?= $d['id_alat'] ?>" class="btn btn-danger btn-sm">Hapus</a>
</td>
</tr>

<?php } ?>
</table>

<?php include '../assets/footer.php'; ?>
```

admin/kategori.php

```php
<?php include '../assets/header.php'; ?>

<?php
if(isset($_POST['simpan'])){
    if($_POST['id']==""){
        mysqli_query($conn,"INSERT INTO kategori VALUES(NULL,'$_POST[nama]','$_POST[deskripsi]')");
    } else {
        mysqli_query($conn,"UPDATE kategori SET
        nama_kategori='$_POST[nama]',
        deskripsi='$_POST[deskripsi]'
        WHERE id_kategori='$_POST[id]'");
    }
}

if(isset($_GET['hapus'])){
    mysqli_query($conn,"DELETE FROM kategori WHERE id_kategori='$_GET[id]'");
}

$edit = null;
if(isset($_GET['edit'])){
    $edit = mysqli_fetch_assoc(mysqli_query($conn,"SELECT * FROM kategori WHERE id_kategori='$_GET[id]'"));
}
?>

<h3>Kelola Kategori</h3>

<form method="post" class="card p-3 mb-3">
<input type="hidden" name="id" value="<?= $edit['id_kategori'] ?? '' ?>">

<input name="nama" class="form-control mb-2" placeholder="Nama Kategori" value="<?= $edit['nama_kategori'] ?? '' ?>">
<input name="deskripsi" class="form-control mb-2" placeholder="Deskripsi" value="<?= $edit['deskripsi'] ?? '' ?>">

<button name="simpan" class="btn btn-success">Simpan</button>
</form>

<table class="table table-bordered">
<tr>
<th>No</th><th>Nama</th><th>Deskripsi</th><th>Aksi</th>
</tr>
```

```php
<?php
$no=1;
$data = mysqli_query($conn,"SELECT * FROM kategori");
while($d=mysqli_fetch_assoc($data)){
?>

<tr>
<td><?= $no++ ?></td>
<td><?= $d['nama_kategori'] ?></td>
<td><?= $d['deskripsi'] ?></td>
<td>
<a href="?edit&id=<?= $d['id_kategori'] ?>" class="btn btn-warning btn-sm">Edit</a>
<a href="?hapus&id=<?= $d['id_kategori'] ?>" class="btn btn-danger btn-sm">Hapus</a>
</td>
</tr>

<?php } ?>
</table>

<?php include '../assets/footer.php'; ?>
```

admin/log.php

```php
<?php include '../assets/header.php'; include '../koneksi.php';

$data=mysqli_query($conn,"
SELECT user.nama, log_aktivitas.aktivitas, log_aktivitas.waktu
FROM log_aktivitas
JOIN user ON log_aktivitas.id_user=user.id_user
ORDER BY log_aktivitas.waktu DESC
");
?>

<h3>Log Aktivitas</h3>

<table class="table table-bordered">
<tr><th>User</th><th>Aktivitas</th><th>Waktu</th></tr>

<?php while($d=mysqli_fetch_assoc($data)){ ?>
<tr>
<td><?= $d['nama'] ?></td>
<td><?= $d['aktivitas'] ?></td>
<td><?= $d['waktu'] ?></td>
</tr>
<?php } ?>
</table>

<?php include '../assets/footer.php'; ?>
```

admin/user.php

```php
<?php include '../assets/header.php'; ?>

<?php
// TAMBAH & EDIT
if(isset($_POST['simpan'])){
    if($_POST['id']==""){
        mysqli_query($conn,"INSERT INTO user VALUES(NULL,'$_POST[nama]','$_POST[username]',md5('$_POST[password]'),'$_POST[role]')");
    } else {
        mysqli_query($conn,"UPDATE user SET
        nama='$_POST[nama]',
        username='$_POST[username]',
        role='$_POST[role]'
        WHERE id_user='$_POST[id]'");
    }
}

// HAPUS
if(isset($_GET['hapus'])){
    mysqli_query($conn,"DELETE FROM user WHERE id_user='$_GET[id]'");
}

// EDIT AMBIL DATA
$edit = null;
if(isset($_GET['edit'])){
    $edit = mysqli_fetch_assoc(mysqli_query($conn,"SELECT * FROM user WHERE id_user='$_GET[id]'"));
}
?>

<h3>Kelola User</h3>

<form method="post" class="card p-3 mb-3">
<input type="hidden" name="id" value="<?= $edit['id_user'] ?? '' ?>">

<input name="nama" class="form-control mb-2" placeholder="Nama" value="<?= $edit['nama'] ?? '' ?>">
<input name="username" class="form-control mb-2" placeholder="Username" value="<?= $edit['username'] ?? '' ?>">

<?php if(!$edit){ ?>
<input name="password" class="form-control mb-2" placeholder="Password">
<?php } ?>
```

```php
<select name="role" class="form-control mb-2">
<option>admin</option>
<option>petugas</option>
<option>peminjam</option>
</select>

<button name="simpan" class="btn btn-primary">Simpan</button>
</form>

<table class="table table-bordered">
<tr>
<th>No</th><th>Nama</th><th>Username</th><th>Role</th><th>Aksi</th>
</tr>

<?php
$no=1;
$data = mysqli_query($conn,"SELECT * FROM user");
while($d=mysqli_fetch_assoc($data)){
?>

<tr>
<td><?= $no++ ?></td>
<td><?= $d['nama'] ?></td>
<td><?= $d['username'] ?></td>
<td><?= $d['role'] ?></td>
<td>
<a href="?edit&id=<?= $d['id_user'] ?>" class="btn btn-warning btn-sm">Edit</a>
<a href="?hapus&id=<?= $d['id_user'] ?>" class="btn btn-danger btn-sm">Hapus</a>
</td>
</tr>

<?php } ?>
</table>

<?php include '../assets/footer.php'; ?>
```

assets/footer.php

```
</div>
```

Assets/header.php

```
<?php
session_start();
include $_SERVER['DOCUMENT_ROOT'].'/peminjaman_alat/koneksi.php';

if(!isset($_SESSION['id'])){
    header("location:/peminjaman_alat/login.php");
}
?>

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
<link rel="stylesheet" href="/peminjaman_alat/assets/style.css">

<div class="sidebar">
    <h4>MENU</h4>

    <?php if($_SESSION['role']=='admin'){ ?>
        <a href="/peminjaman_alat/dashboard.php">Dashboard</a>
        <a href="/peminjaman_alat/admin/user.php">User</a>
        <a href="/peminjaman_alat/admin/kategori.php">Kategori</a>
        <a href="/peminjaman_alat/admin/alat.php">Alat</a>
        <a href="/peminjaman_alat/admin/log.php">Log</a>
    <?php } ?>

    <?php if($_SESSION['role']=='petugas'){ ?>
        <a href="/peminjaman_alat/dashboard.php">Dashboard</a>
        <a href="/peminjaman_alat/petugas/peminjaman.php">ACC Peminjaman</a>
        <a href="/peminjaman_alat/petugas/pengembalian.php">Pengembalian</a>
        <a href="/peminjaman_alat/petugas/laporan.php">Laporan</a>
    <?php } ?>

    <?php if($_SESSION['role']=='peminjam'){ ?>
        <a href="/peminjaman_alat/dashboard.php">Dashboard</a>
        <a href="/peminjaman_alat/user/alat.php">Lihat Alat</a>
        <a href="/peminjaman_alat/user/pinjam.php">Pinjam</a>
        <a href="/peminjaman_alat/user/kembali.php">Pengembalian</a>
    <?php } ?>

    <a href="/peminjaman_alat/logout.php">Logout</a>
</div>

<div class="content">
```

Assets/style.css

```
body {
    background: #c4e1ff;
    font-family: 'Segoe UI';
}

.sidebar {
    width: 220px;
    height: 100vh;
    position: fixed;
    background: #1e3a8a;
    color: white;
    padding: 20px;
}

.sidebar a {
    display: block;
    color: white;
    padding: 10px;
    text-decoration: none;
    border-radius: 8px;
    margin-bottom: 5px;
}

.sidebar a:hover {
    background: #3b82f6;
}

.content {
    margin-left: 240px;
    padding: 20px;
}

.card {
    border-radius: 15px;
}
```

Petugas/acc.php

```php
<?php include '../assets/header.php'; include '../koneksi.php'; ?>

<h3>ACC Peminjaman</h3>

<table class="table">
<tr><th>User</th><th>Status</th><th>Aksi</th></tr>

<?php
$data=mysqli_query($conn,"
SELECT peminjaman.*, user.nama
FROM peminjaman
JOIN user ON peminjaman.id_user=user.id_user
WHERE status='dipinjam'
");

while($d=mysqli_fetch_assoc($data)){
?>
<tr>
<td><?= $d['nama'] ?></td>
<td><?= $d['status'] ?></td>
<td><button class="btn btn-success btn-sm">ACC</button></td>
</tr>
<?php } ?>
</table>

<?php include '../assets/footer.php'; ?>
```

Petugas/cetak.php

```php
<?php
include '../koneksi.php';
?>

<h3>Laporan Peminjaman</h3>

<table border="1" width="100%" cellpadding="5">
<tr>
<th>No</th><th>User</th><th>Alat</th><th>Pinjam</th><th>Kembali</th><th>Status</th>
</tr>

<?php
$no=1;
$data = mysqli_query($conn,"
SELECT p.*, u.nama, a.nama_alat
FROM peminjaman p
JOIN user u ON p.id_user=u.id_user
JOIN detail_peminjaman d ON p.id_peminjaman=d.id_peminjaman
JOIN alat a ON d.id_alat=a.id_alat
");

while($d=mysqli_fetch_assoc($data)){
?>

<tr>
<td><?= $no++ ?></td>
<td><?= $d['nama'] ?></td>
<td><?= $d['nama_alat'] ?></td>
<td><?= $d['tanggal_pinjam'] ?></td>
<td><?= $d['tanggal_kembali'] ?></td>
<td><?= $d['status'] ?></td>
</tr>

<?php } ?>
</table>

<script>
window.print();
</script>
```

Petugas/laporan.php

```php
<?php include '../assets/header.php'; ?>

<h3>Laporan Peminjaman</h3>

<a href="cetak.php" target="_blank" class="btn btn-success mb-3">Cetak Laporan</a>

<table class="table table-bordered">
<tr>
<th>No</th><th>User</th><th>Alat</th><th>Pinjam</th><th>Kembali</th><th>Status</th>
</tr>

<?php
$no=1;
$data = mysqli_query($conn,"
SELECT p.*, u.nama, a.nama_alat
FROM peminjaman p
JOIN user u ON p.id_user=u.id_user
JOIN detail_peminjaman d ON p.id_peminjaman=d.id_peminjaman
JOIN alat a ON d.id_alat=a.id_alat
");

while($d=mysqli_fetch_assoc($data)){
?>

<tr>
<td><?= $no++ ?></td>
<td><?= $d['nama'] ?></td>
<td><?= $d['nama_alat'] ?></td>
<td><?= $d['tanggal_pinjam'] ?></td>
<td><?= $d['tanggal_kembali'] ?></td>
<td><?= $d['status'] ?></td>
</tr>

<?php } ?>
</table>

<?php include '../assets/footer.php'; ?>
```

Petugas/peminjaman.php

```php
<?php include '../assets/header.php'; ?>

<?php
// ACC
if(isset($_GET['acc'])){
    $id = $_GET['id'];

    mysqli_query($conn,"UPDATE peminjaman SET status='dipinjam' WHERE id_peminjaman='$id'");

    // ambil detail
    $d = mysqli_fetch_assoc(mysqli_query($conn,"
    SELECT * FROM detail_peminjaman WHERE id_peminjaman='$id'
    "));

    // kurangi stok
    mysqli_query($conn,"
    UPDATE alat SET stok = stok - $d[jumlah]
    WHERE id_alat='$d[id_alat]'
    ");

    // LOG
    mysqli_query($conn,"
    INSERT INTO log_aktivitas (id_user,aktivitas)
    VALUES ('".$_SESSION['id']."','ACC peminjaman')
    ");
}
?>

<h3>ACC Peminjaman</h3>

<table class="table table-bordered">
<tr>
<th>No</th><th>User</th><th>Tanggal</th><th>Status</th><th>Aksi</th>
</tr>
```

```php
<?php
$no=1;
$data = mysqli_query($conn,"
SELECT peminjaman.*, user.nama
FROM peminjaman
JOIN user ON peminjaman.id_user=user.id_user
WHERE status='menunggu'
");

while($d=mysqli_fetch_assoc($data)){
?>

<tr>
<td><?= $no++ ?></td>
<td><?= $d['nama'] ?></td>
<td><?= $d['tanggal_pinjam'] ?></td>
<td><?= $d['status'] ?></td>
<td>
<a href="?acc&id=<?= $d['id_peminjaman'] ?>" class="btn btn-success btn-sm">ACC</a>
</td>
</tr>

<?php } ?>
</table>

<?php include '../assets/footer.php'; ?>
```

Petugas/pengembalian.php

```php
<?php include '../assets/header.php'; ?>

<?php
// PROSES KEMBALIKAN
if(isset($_GET['kembali'])){
    $id = $_GET['id'];

    // ambil data peminjaman
    $pinjam = mysqli_fetch_assoc(mysqli_query($conn,"
    SELECT * FROM peminjaman WHERE id_peminjaman='$id'
    "));

    $tgl_kembali = $pinjam['tanggal_kembali'];
    $hari_ini = date('Y-m-d');

    // HITUNG DENDA (misal 1000/hari telat)
    $denda = 0;
    if($hari_ini > $tgl_kembali){
        $telat = (strtotime($hari_ini) - strtotime($tgl_kembali)) / (60*60*24);
        $denda = $telat * 1000;
    }

    // simpan ke tabel pengembalian
    mysqli_query($conn,"
    INSERT INTO pengembalian (id_peminjaman,tanggal_dikembalikan,denda)
    VALUES ('$id','$hari_ini','$denda')
    ");

    // update status
    mysqli_query($conn,"
    UPDATE peminjaman SET status='selesai'
    WHERE id_peminjaman='$id'
    ");

    // ambil detail alat
    $detail = mysqli_fetch_assoc(mysqli_query($conn,"
    SELECT * FROM detail_peminjaman WHERE id_peminjaman='$id'
    "));

    // kembalikan stok
    mysqli_query($conn,"
    UPDATE alat SET stok = stok + $detail[jumlah]
    WHERE id_alat='$detail[id_alat]'
    ");
```